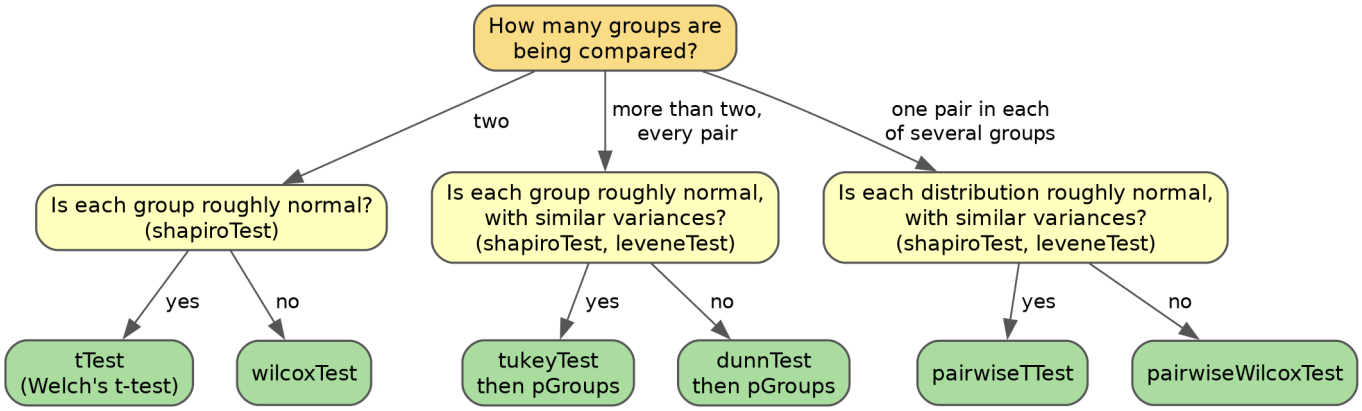
comparing means: summary



Before choosing a statistical test to apply to your data, look at the distribution of the data in each group. **Can each group be modeled by a normal distribution?** Check with the Shapiro test within each group. If they are not significantly different from normal, use a parametric test. If they are, use a rank test, which asks whether values in one group tend to be higher than values in the other. **How many groups are being compared?** With more than two, a separate t-test on every pair multiplies the chance of a false positive, so the Tukey and Dunn tests compare every pair at once and correct the p-values (read `p.adj`). The t-test used in this book is Welch's, so two groups need no variance test, but the Tukey test shares one estimate of the spread across groups, so check for equal variances first using the Levene test. The *p* value is the likelihood of getting data as extreme as the data in hand if the distributions they are coming from are in fact the same. The *p* value is not the probability that a detected difference is a false positive. Choosing a test comes down to how many groups are compared and whether the data are normal. Note that ANOVA is addressed in a separate, dedicated chapter.

Function	What it is for	How it is used in this chapter
`shapiroTest(abundance)`	is each group roughly normal? (p > 0.05 means not significantly different from normal)	after `group_by()` on the grouping column(s)
`leveneTest(abundance ~ aquifer_code)`	do the groups have similar variances?	before `tukeyTest` or `pairwiseTTest`
`tTest(abundance ~ aquifer_code)`	two groups, roughly normal (Welch's t-test)	`y ~ x` formula
`wilcoxTest(abundance ~ aquifer_code)`	two groups, not normal (rank test)	`y ~ x` formula
`tukeyTest(abundance ~ aquifer_code)`	more than two groups, every pair, normal with similar variances	read `p.adj`
`dunnTest(abundance ~ aquifer_code)`	more than two groups, every pair, rank-based	read `p.adj`
`pGroups()`	letters for each group from Tukey or Dunn output (a shared letter means no significant difference)	piped after `tukeyTest` or `dunnTest`, then drawn with `geom_text()`
`pairwiseTTest(abundance ~ analyte)`, `pairwiseWilcoxTest(abundance ~ analyte)`	one comparison within each of several groups	after `group_by(aquifer_code)`, with `p.adj` corrected across all the groups
`geomSignif()`	draws brackets and stars from pairwise test output	`data =` a table built from the test output